(Please write your Enrollment Number) Enrollment No._____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Off Line)
Course Name: Machine Learning, Semester:5th
(Nov-Dec, 2022)

Subject Code: BAI 301	Subject: Machine Learning
Time :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5×8=20)

(a) List 5 areas where machine learning can not possibly have applications.
(b) What is the motivation of doing away with merely train-test split in favor of additional validation dataset.
(c) Draw two examples of non-linearly separable datasets.
(d) Why is kNN called lazy learner?
(e) What is unclassified in unsupervised machine learning context?
(f) Give two real-life use cases where partitional clustering is relevant.
(g) Name any 3 metrics for evaluating learned hypotheses and any 2 metrics for comparing learning algorithms.
(h) What is adapted and what is boosted in adaptive boosting?

UNIT-I

Q2 Identify different data attribute types and their similarity measures. (10)

Q3 Illustrate k-fold cross validation with a dataset of size 100 and k=5. (10)

UNIT-II

Q4 List one merit and one demerit each of logistic classifier, support vector classifier, naive Bayes classifier, k nearest neighbor and decision tree classifier. (10)

Q5 Give a case in point for regularization-what, why, and how(any 2 techniques), starting with linear regression drawbacks. (10)

UNIT-III

Q6 Cluster the following eight points (with (x, y) representing locations) into three clusters: (10)

A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Q7 Cluster the following 5 data points using agglomerative clustering and show dendogram. Use any linkage measure,specify the same. (10)

	1	2	3	4	5
1	0				
2	9	0			
3	3	7	0		
4	6	5	9	0	
5	11	10	2	8	0

PTO

UNIT-IV

Q8 For the following dataset, plot ROC curve and find AUC at thresholds- 0.0,0.7,0.8,0.9,1.0. **(10)**

index	class	probability
P1	1	0.95
P2	1	0.90
P3	0	0.85
P4	0	0.81
P5	1	0.78
P6	1	0.70

Q9 Contrast bagging and boosting with figures. **(10)**